Network layer

Network layer

- Network Layer is the third layer from the bottom (Layer 3) and the fifth layer from the top in the OSI (Open Systems Interconnection) Model. It is responsible for ensuring end-to-end packet delivery across multiple interconnected networks.
- ***Note:*** *Unlike the Data Link Layer, which focuses only on node-to-node delivery within a single network segment, the Network Layer ensures that data travels from the source host to the destination host, even if they are located on different networks*

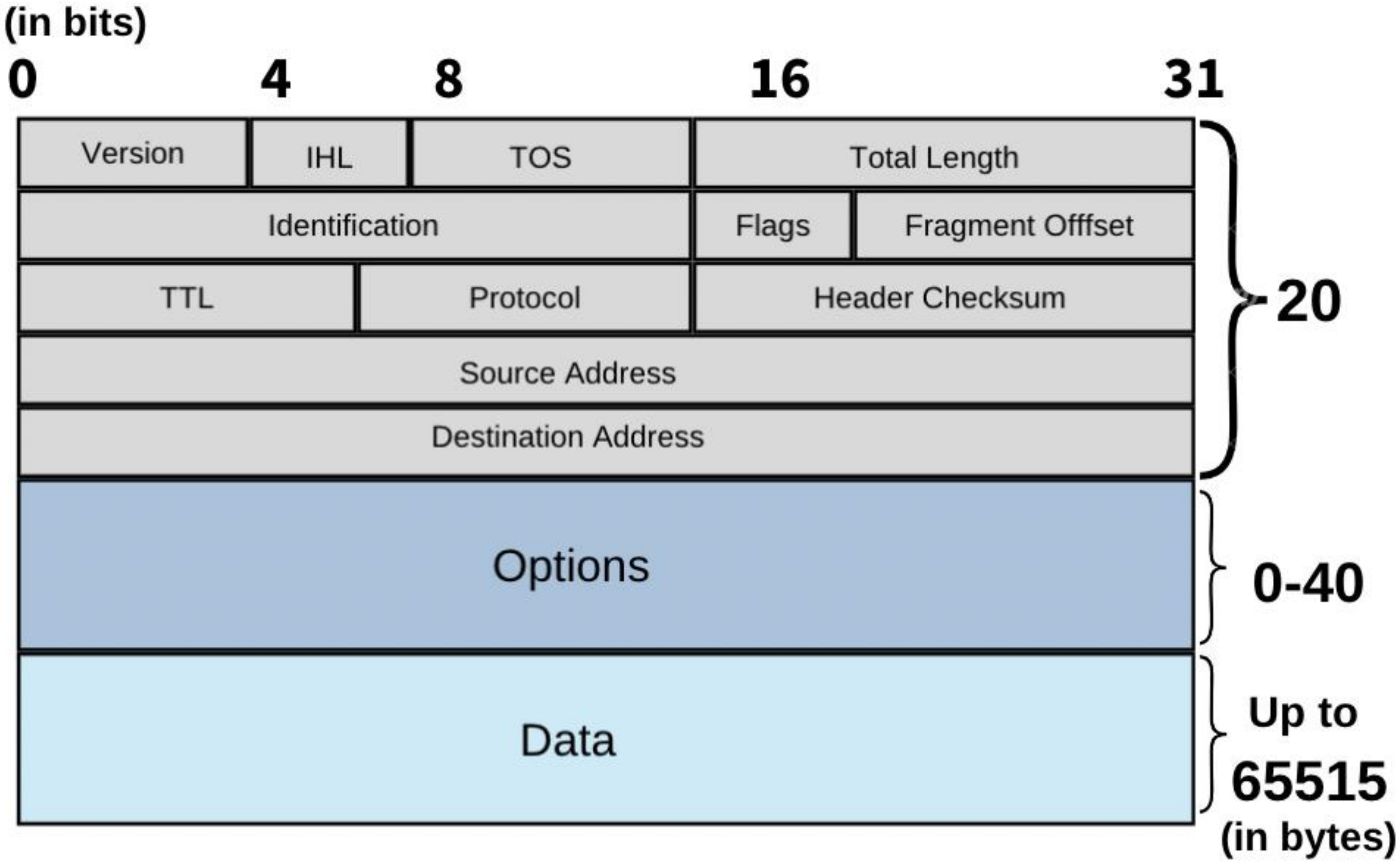
Functions of network layer

Function	Description
Logical Addressing	Assigns unique IP addresses to devices to identify them on the network (e.g., IPv4 or IPv6).
Routing	Determines the best path for data to travel from source to destination using routing algorithms.
Packet Forwarding	Moves packets from one network to another using routers.
Fragmentation and Reassembly	Divides large packets into smaller fragments if the next network cannot handle large sizes. Reassembles them at the destination.
Error Handling and Diagnostics	Detects and manages errors in routing (e.g., unreachable destination). Tools like ICMP (ping, traceroute) are used.
Congestion Control	Manages traffic to avoid network overload.

- Data is in packet.
- A **packet** is the **basic unit of data** used by the **Network Layer (Layer 3)** in the OSI model.
  It carries data from the **source device to the destination device** across multiple networks using **logical addressing (IP addresses).**
-

  A packet = Header + Payload (data) + Trailer (optional)

Structure of network layer packet



Field	Size (bits)	Description
Version	4	Indicates IP version (IPv4 or IPv6).
Header Length (IHL)	4	Tells how long the header is.
Type of Service (ToS)	8	Defines priority or QoS level.
Total Length	16	Total size of the packet (header + data).
Identification	16	Used for identifying fragments of a single packet.
Flags	3	Controls fragmentation (e.g., don't fragment).
Fragment Offset	13	Used to reassemble fragmented packets.
Time To Live (TTL)	8	Limits packet lifetime to prevent loops.
Protocol	8	Identifies the upper-layer protocol (e.g., TCP = 6, UDP = 17).
Header Checksum	16	Error checking for the header.
Source IP Address	32	Address of sender.
Destination IP Address	32	Address of receiver.
Options	Variable	Used for testing, debugging, or control (optional).
Data (Payload)	Variable	The actual message/data from upper layers.

Switching

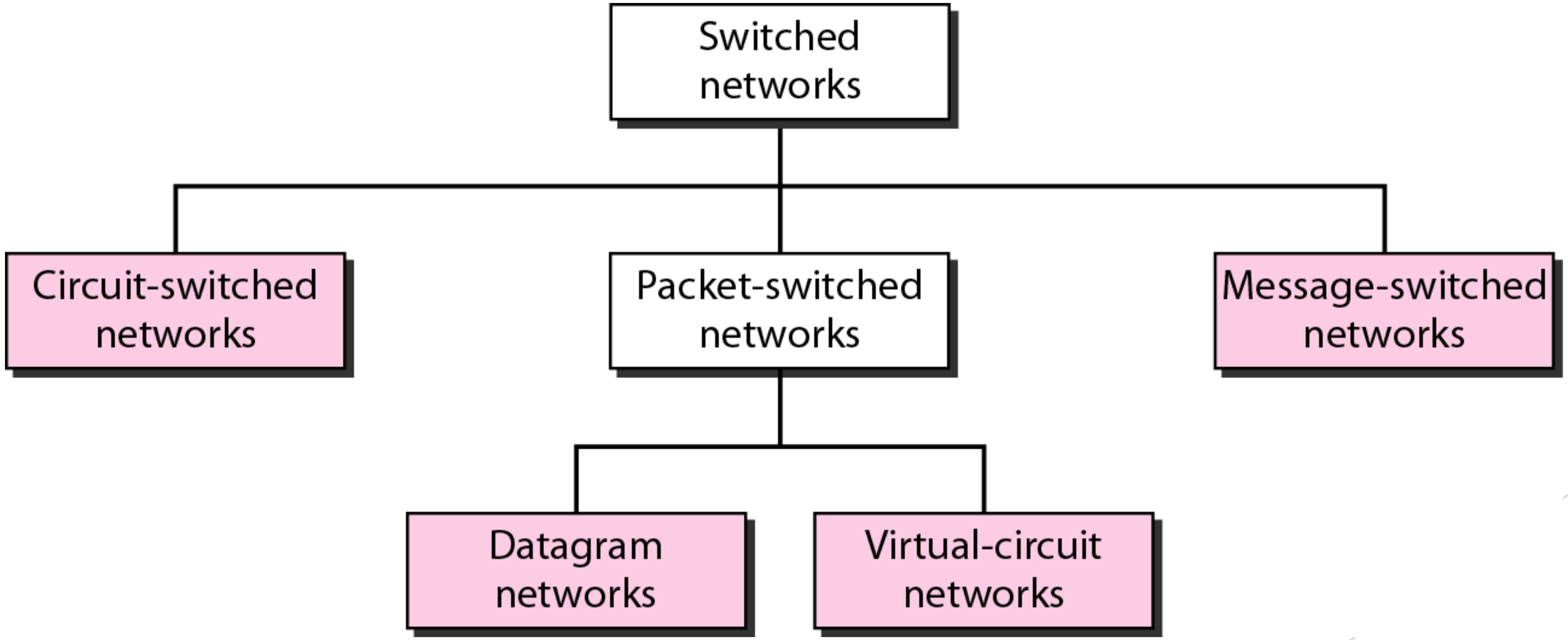
- **Switching** is the **process of moving data packets** from the **source device** to the **destination device** through a **network of intermediate nodes (switches or routers)**

Purpose of switching

- To **connect multiple devices or networks**.
- To **choose the best path** for packet delivery.
- To **forward data efficiently** between sender and receiver.
- To **reduce congestion** and **improve network performance**.

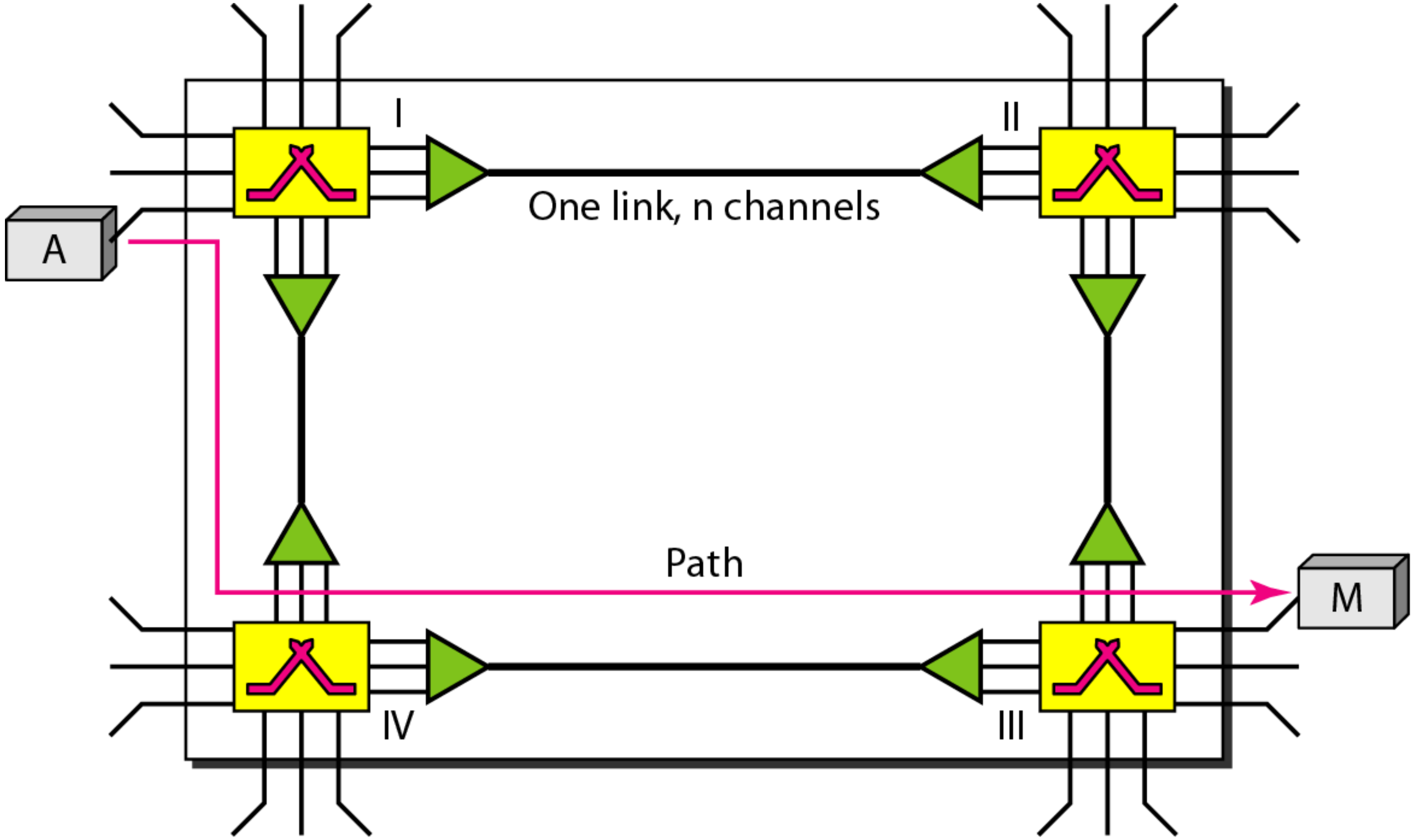
Types of switching

- Circuit switch
- Packet switch
- Message switch



Circuit Switching

- Circuit Switching establishes a dedicated path between sender and receiver before communication, using the full network bandwidth.
- it is costly and inefficient for high-traffic or large networks due to reserved resources.
- A circuit-switched network consists of a set of switches connected by physical links.
- The telephone system network is one of the examples of Circuit switching.
- FDM (Frequency Division Multiplexing) and TDM (Time Division Multiplexing) are two methods of multiplexing multiple signals into a single carrier.


I
II
One link, n channels
A
Path
M
IV
III

FDM (Frequency Division Multiplexing)

- Frequency Division Multiplexing (FDM) is a technique that allows multiple signals to be transmitted simultaneously over a single communication channel by assigning each signal a different frequency band within the available bandwidth.

Time Division Multiplexing (TDM)

- Time Division Multiplexing (TDM) is a technique that allows multiple signals to share the same transmission channel by dividing the available time into time slots, each assigned to a different signal.

Phases of Circuit Switching

- **Circuit Establishment:** A dedicated circuit between the source and destination is constructed via a number of intermediary switching center's
- **Data Transfer:** Data can be transferred between the source and destination once the circuit has been established
- **Circuit Disconnection:** Disconnection in the circuit occurs when one of the users initiates the disconnect. When the disconnection occurs, all intermediary linkages between the sender and receiver are terminated.

Advantages

- **Guaranteed data rate:** Dedicated channel ensures consistent transmission speed.
- **No delay in data flow:** Dedicated path allows continuous data transfer.
- **High reliability:** Reserved path prevents data loss or corruption.

Disadvantages

- **Limited scalability:** Not suitable for large networks; needs a dedicated path for each communication.
- **Vulnerable to failures:** If a path fails, communication is disrupted.
- **Limited flexibility:** Dedicated circuits cannot be used by others until communication ends.
- **Wastes resources:** Bandwidth is reserved even when no data is sent.
- **High cost:** Setting up and maintaining dedicated paths is expensive.

Total Delay in Circuit Switching

- **Circuit Establishment time** $\rightarrow T_s$
- **Data Transmission time** $\rightarrow T_t$
- **Circuit Teardown (release) time** $\rightarrow T_d$
- Total time taken to transmit a message in circuit switched network

= Connection set up time + Transmission delay + Propagation delay + Tear down time

- Transmission delay = Message size / Bandwidth
- Propagation delay = (Number of hops on way x Distance between 2 hops) / Propagation speed

Efficiency of Circuit Switching

- efficiency depends on **how much of that time is actually used to send data.**
- If total connection time = T_{total}
  and actual data transfer time = T_t
- Efficiency = (Data Transfer Time ÷ Total Connection Time) × 100 percent

Problem

- Consider all links in the network use TDM with 24 slots and have a data rate of 1.536 Mbps. Assume that host A takes 500 msec to establish an end to end circuit with host B before begin to transmit the file. If the file is 512 kilobytes, then how much time will it take to send the file from host A to host B?

solution

- Given-
- Total bandwidth = 1.536 Mbps
- Bandwidth is shared among 24 slots
- Connection set up time = 500 msec
- File size = 512 KB

Calculating Bandwidth Per User-

- Total bandwidth = Number of users x Bandwidth per user
- So, Bandwidth per user
- = Total bandwidth / Number of users
- = 1.536 Mbps / 24
- = 0.064 Mbps
- = 64 Kbps

Calculating Transmission Delay-

- Transmission delay (T_t)
- = File size / Bandwidth
- = 512 KB / 64 Kbps
- = $(512 \times 2^{10} \times 8$ bits$) / (64 \times 10^3$ bits per sec$)$
- = 65.536 sec
- = 65536 msec

Calculating Time Required To Send File-

- = Connection set up time + Transmission delay
- = 500 msec + 65536 msec
- = 66036 sec
- = 66.036 msec

Message Switching

- Message Switching is a switching technique where the entire message is sent as a single unit from one node to another without a dedicated path between the sender and receiver.
- Each intermediate node stores the message temporarily and then forwards it to the next node until it reaches its destination.

Key Features of Message Switching

Store-and-Forward

- Each intermediate node must store the entire message before forwarding it.
- Transmission occurs only if the next link and node are available; otherwise, the message waits in storage until resources are free.

Message Delivery

- Messages are transmitted as complete units, with a header containing routing information (source and destination addresses).
- Communication is hop by hop until the message reaches its destination.

Advantage

- As message switching is able to store the message for which communication channel is not available, it helps in reducing the traffic congestion in the network.
- In message switching, the data channels are shared by the network devices.
- It makes traffic management efficient by assigning priorities to the messages.
- It allows for infinite message lengths

Disadvantage

- **Not suitable for real-time applications:** Storing messages at intermediate nodes causes delays.
- **High storage requirement:** Every intermediate node needs large storage capacity.
- **Delivery uncertainty:** Complex system can make it unclear if messages are correctly delivered.
- **No dedicated path:** Communication is less reliable because sender and receiver are not directly connected.

Problem

- **Message size (D)** = 120,000 bytes = 120,000 × 8 = **960,000 bits**
- **Link bandwidth (B)** = 1 Mbps = **1,000,000 bits per second**
- **Propagation delay per link (Tp)** = **10 ms = 0.01 seconds**
- **Processing delay at each intermediate switch (Tproc)** = **5 ms = 0.005 seconds**
- **Number of links (L)** = 3 (Source → R1 → R2 → Destination)

→ So, **2 intermediate nodes (R1 and R2)**

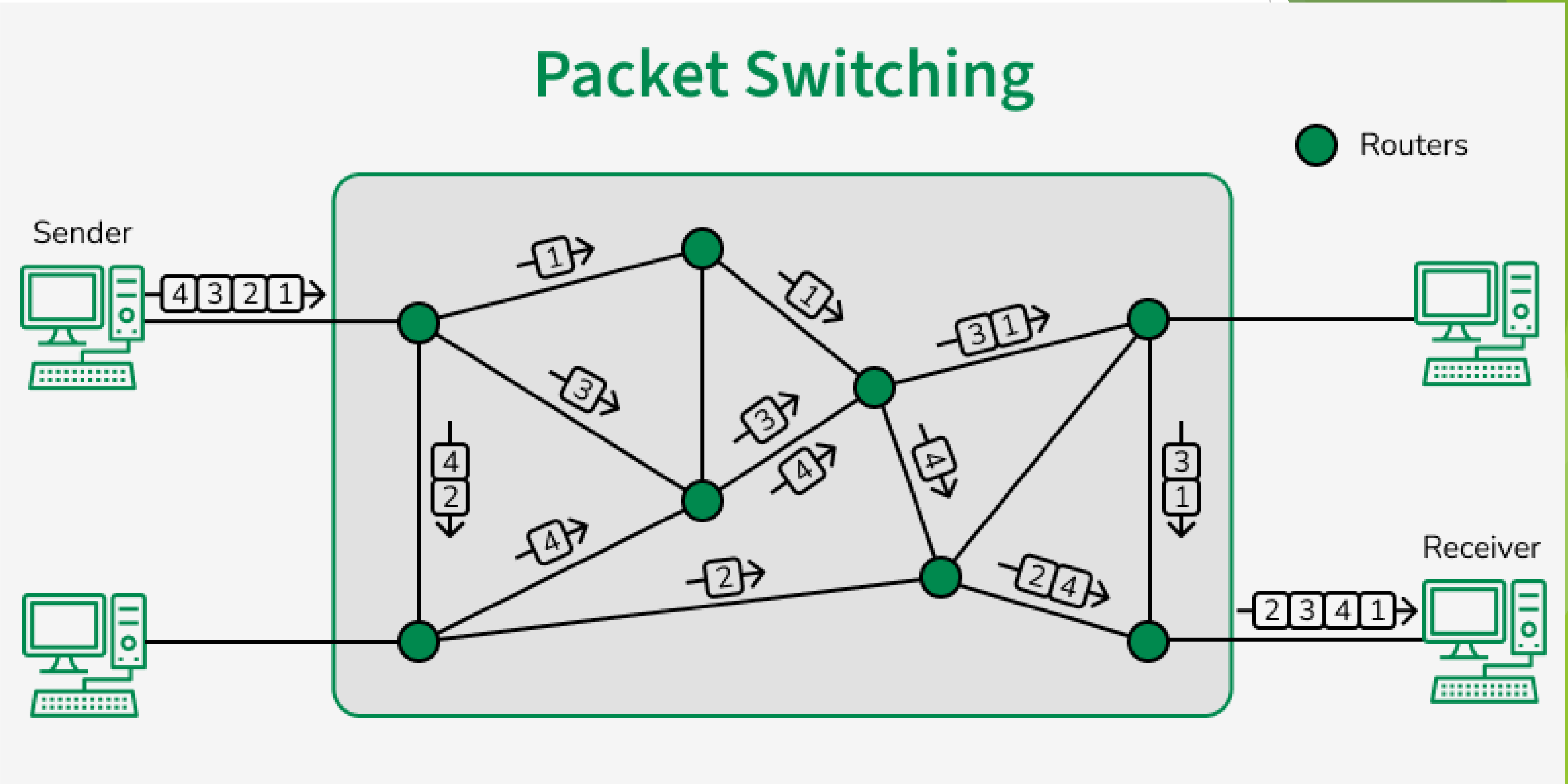
- **Transmission time on one link**
- $T_{tx} = \frac{D}{B} = \frac{960{,}000}{1{,}000{,}000} = 0.96\ seconds$
-

- **Step 2 — Total time per link (Transmission + Propagation)**
- $T_{per\_link} = T_{tx} + T_p = 0.96 + 0.01 = 0.97\ seconds$

- Total time across all links (Store-and-Forward)
- So, total time includes all links' transmission + propagation + processing delays:
- $T_{links} = L \times T_{per\_link} = 3 \times 0.97 = 2.91\ \mathit{seconds}$
- $T_{processing\_total} = (L-1) \times T_{proc} = 2 \times 0.005 = 0.01\ \mathit{seconds}$
- Therefore:
- $\boxed{T_{total} = T_{links} + T_{processing\_total} = 2.91 + 0.01 = 2.92\ \mathit{seconds}}$

Packet Switching

- Packet switching is a technique used in computer networks (like the Internet) where data is broken into packets before being sent to the destination. Each packet may take a **different route**, and all are **reassembled** in the correct order at the destination..
- No dedicated path or resource reservation is needed, making data transfer efficient and flexible.


Packet Switching
Routers
Sender
4 3 2 1
Receiver
2 3 4 1

Need for Packet Switching

- Earlier networks used **Circuit Switching** (like telephone lines), where a **dedicated path** was established for communication.
- This caused **inefficient use of bandwidth**, because even when no data was sent, the channel stayed reserved.
- To solve this, packet switching allows **multiple users to share the same network resources**, increasing efficiency.

Working Process

- Message Division → The original message is split into packets (e.g., 1 MB file divided into 1000 packets of 1 KB each).
- Addressing → Each packet gets a header with source & destination addresses and sequence numbers.
- Routing → Each packet travels independently through the network; routers choose the best available path using routing algorithms.
- Arrival at Destination → Packets may arrive out of order, through different routes, or even at different times.
- Reassembly → The receiving device arranges all packets according to their sequence numbers to rebuild the original message.
- Error Checking → If some packets are missing or damaged, the receiver requests retransmission

Advantages

- **Efficient use of bandwidth** – No dedicated path; bandwidth is shared among many users.
- **High reliability** – If one route fails, packets are automatically rerouted through another path.
- **Resource sharing** – Many users can send data at the same time using the same network.
- **Scalability** – Easy to add new devices and expand the network
- **Better utilization during idle times** – Bandwidth not used by one user can be used by others.

- **Packet loss possible** – Routers may drop packets when the network is congested.
- **Out-of-order delivery** – Packets can arrive in a different order and must be rearranged.
- **Complex reassembly process** – Receiver must reassemble packets in correct order.
- **High processing load** – Routers must inspect and route each packet individually.

Types of packet switching

- **Virtual Circuit**
- Connection-oriented (Virtual Circuit) Packet Switching establishes a logical path between sender and receiver before data transfer. All packets follow this predefined route and are given sequence numbers to ensure they arrive in order
- This method involves three main phases:
- **1. Setup Phase**: A path is established between sender and receiver.
- **Data Transfer Phase**: Packets are transmitted along the established route
- **Tear Down Phase**: After the transmission is complete, the virtual circuit is released.

Datagram packet switching

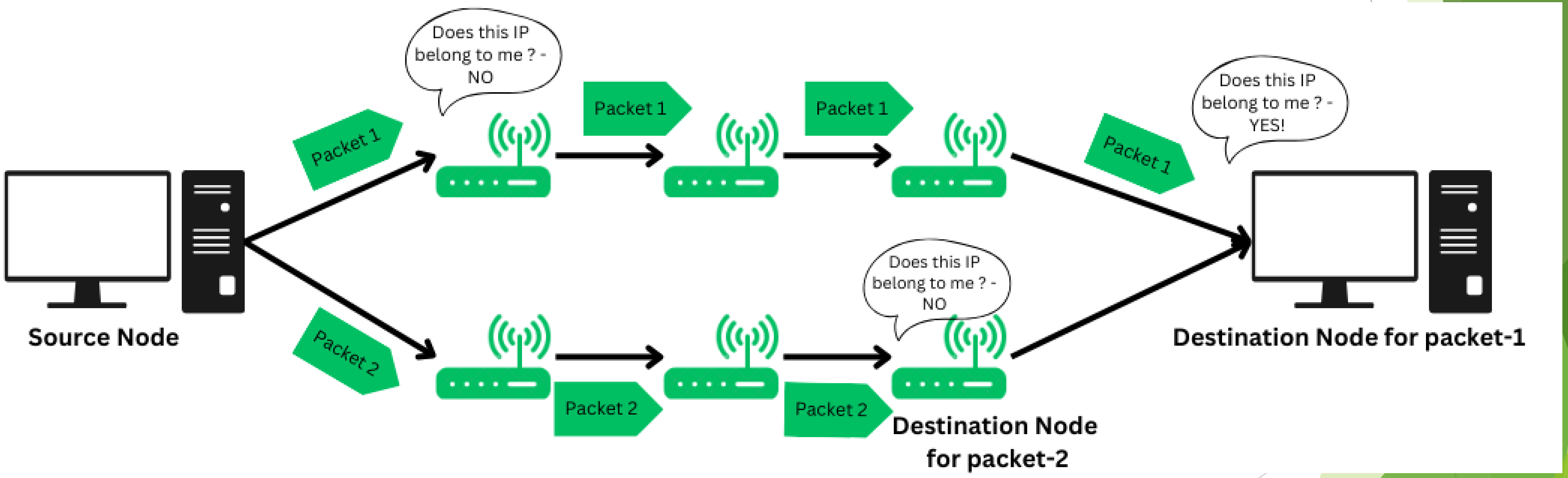
- Connectionless (Datagram) Packet Switching treats each packet independently, with all addressing and control information included. No connection setup is needed, and packets may take different paths, possibly arriving out of order. Reliability and data integrity are handled by higher-layer protocols like TCP, providing flexibility and speed.

Problem assignment

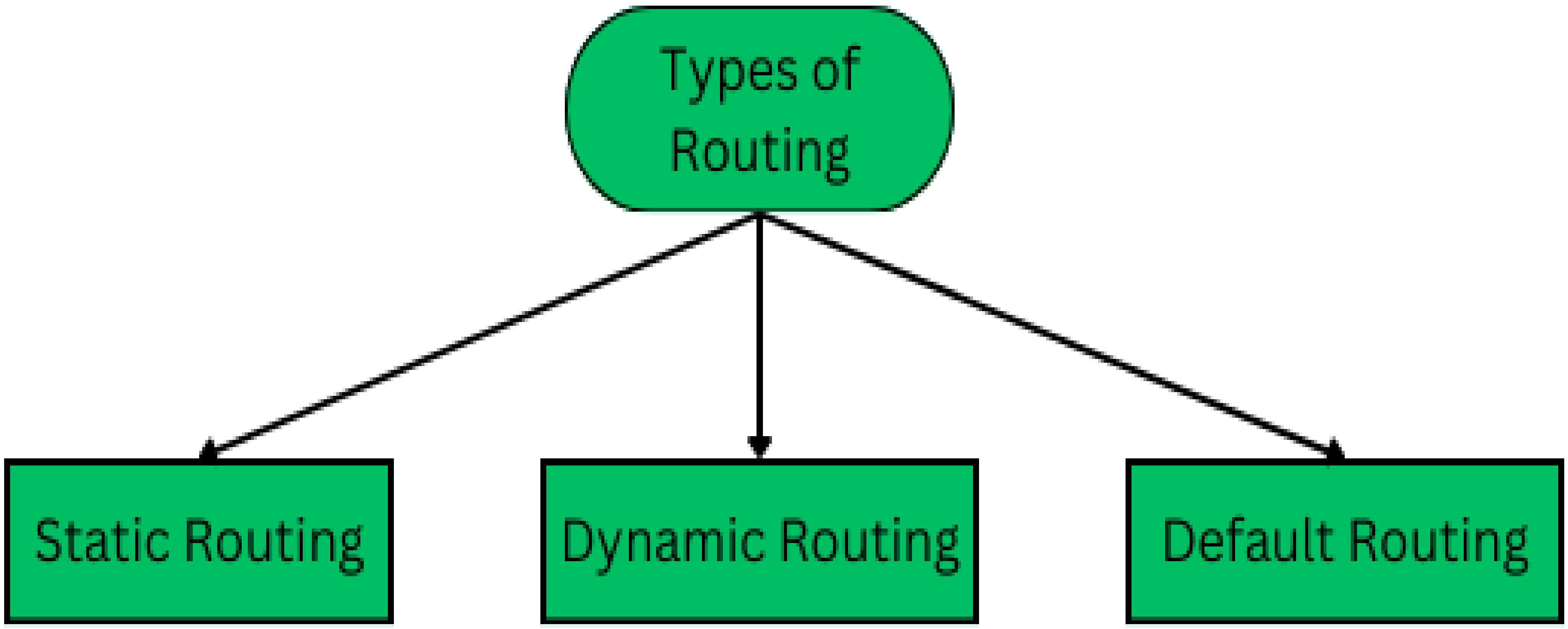
- There is a network having bandwidth of 1 MBps.
- A message of size 1000 bytes has to be sent.
- Packet switching technique is used.
- Each packet contains a header of 100 bytes.
- Out of the following, in how many packets the message must be divided so that total time taken is minimum-
- 1 packet
- 5 packets
- 10 packets
- 20 packets

ROUTING

- The process of choosing a path across one or more networks is known as Network Routing.

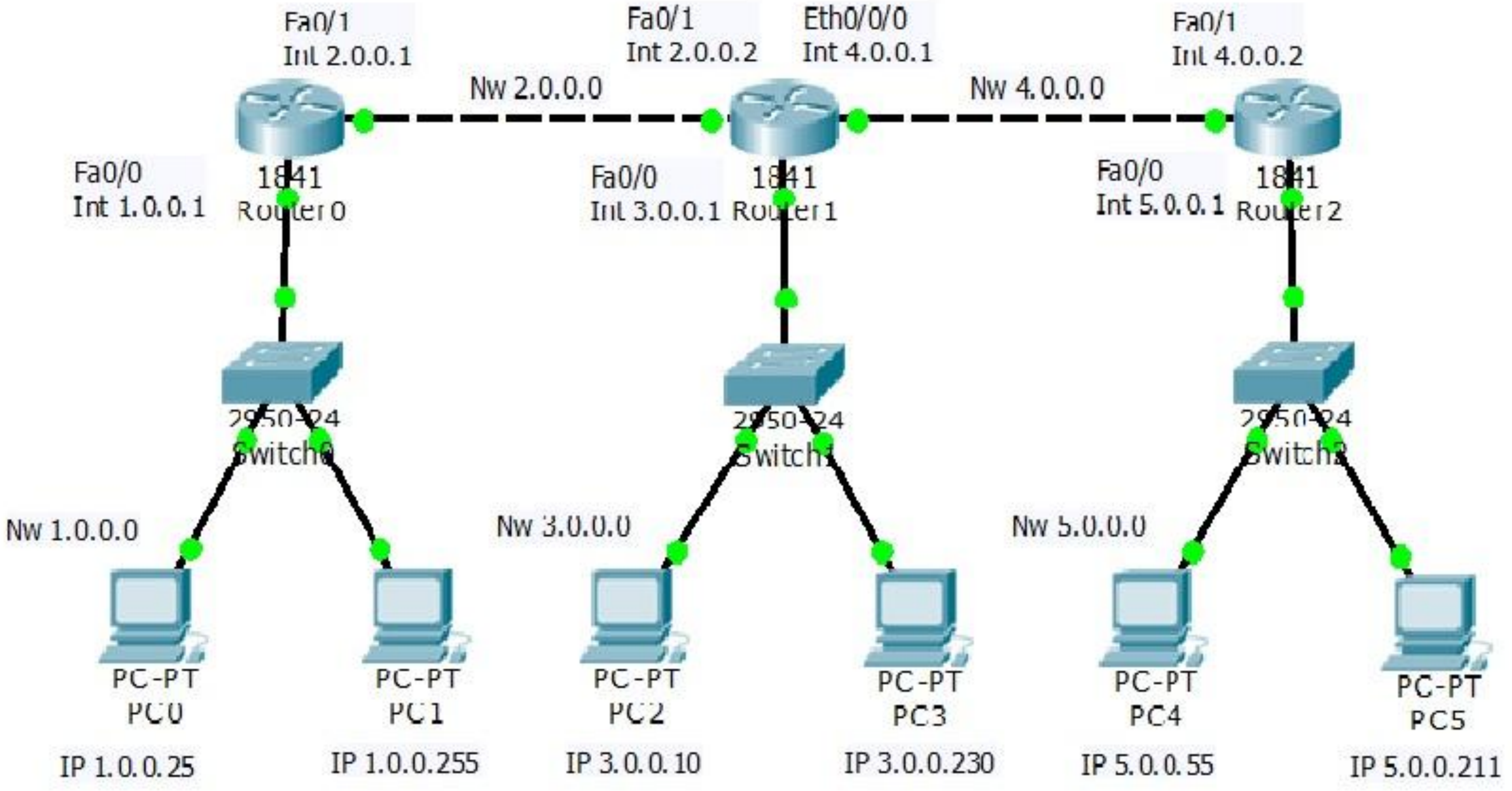


- The Source Node (Sender) sends the data packet on the network, embedding the IP in the header of the data packet.
- The nearest router receives the data packet, and based on some metrics, further routes the data packet to other routers.
- Step 2 occurs recursively till the data packet reaches its intended destination.


Types of Routing
Static Routing
Dynamic Routing
Default Routing

Static routing

- Static routing is also called as "non-adaptive routing". In this, routing configuration is done manually by the network administrator. Let's say for example, we have 5 different routes to transmit data from one node to another, so the network administrator will have to manually enter the routing information by assessing all the routes.
- A network administrator has full control over the network, routing the data packets to their concerned destinations
- Routers will route packets to the destination configured manually by the network administrator.
- Although this type of routing gives fine-grained control over the routes, it may not be suitable for large-scale enterprise networks.
- A real-time example of static routing is a small office network where a single router connects the local office network to the internet


Fa0/1
Int 2.0.0.1
Nw 2.0.0.0
Fa0/1
Int 2.0.0.2
Eth0/0/0
Int 4.0.0.1
Nw 4.0.0.0
Fa0/1
Int 4.0.0.2
Fa0/0
Int 1.0.0.1
1841
Router0
Fa0/0
Int 3.0.0.1
1841
Router1
Fa0/0
Int 5.0.0.1
1841
Router2
2950-24
Switch0
2950-24
Switch1
2950-24
Switch2
Nw 1.0.0.0
Nw 3.0.0.0
Nw 5.0.0.0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
IP 1.0.0.25
IP 1.0.0.255
IP 3.0.0.10
IP 3.0.0.230
IP 5.0.0.55
IP 5.0.0.211

Advantages

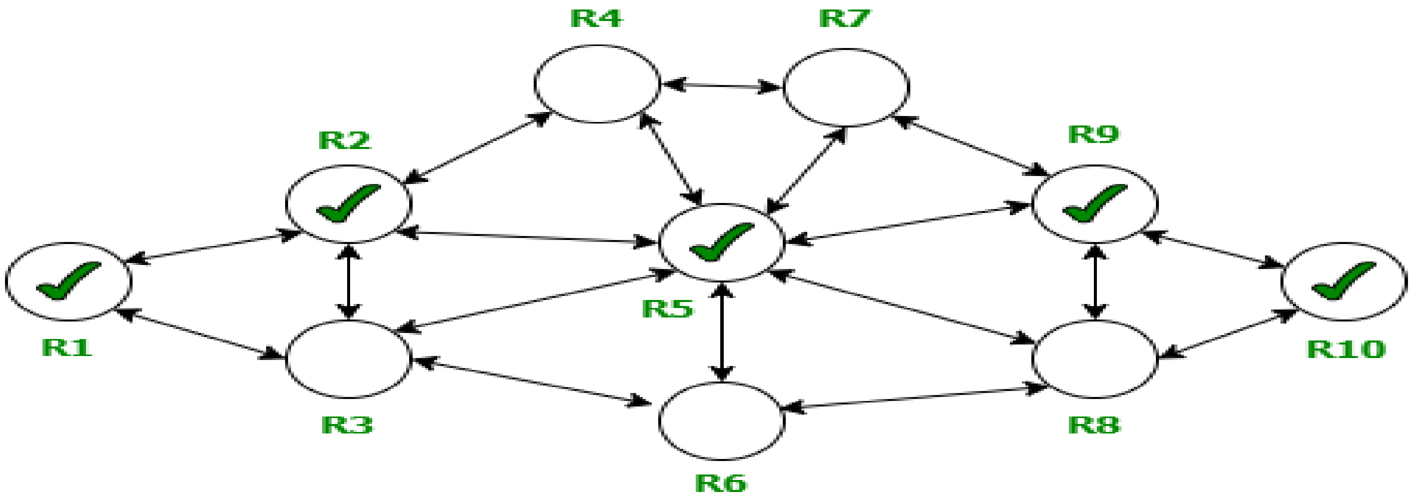
- **High security:** Since routes are manually defined, it prevents unauthorized routing to certain networks.
- **Predictable performance:** The path is always the same, leading to predictable and consistent traffic flow.

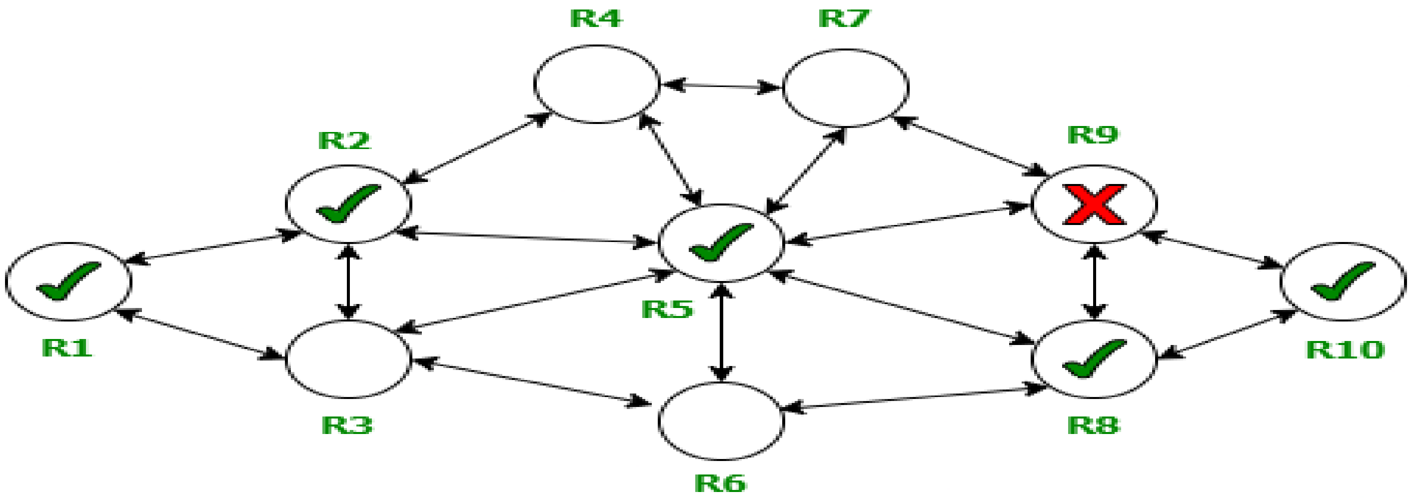
Disadvantages

- **Not scalable:** It is not practical for large or complex networks, as manual updates cannot keep up with frequent changes.
- **Inflexible:** It cannot automatically reroute traffic around a link failure without manual intervention, leading to a lack of automatic failover.
- **Prone to human error:** Manual configuration is susceptible to human mistakes, such as incorrect IP addresses or subnet masks
- It is a burdensome task to sum up or add-on each route manually to the routing map in a large network.
- Managing its ordering is time-consuming.
- It cannot reroute traffic in case some link fails.

Dynamic Routing

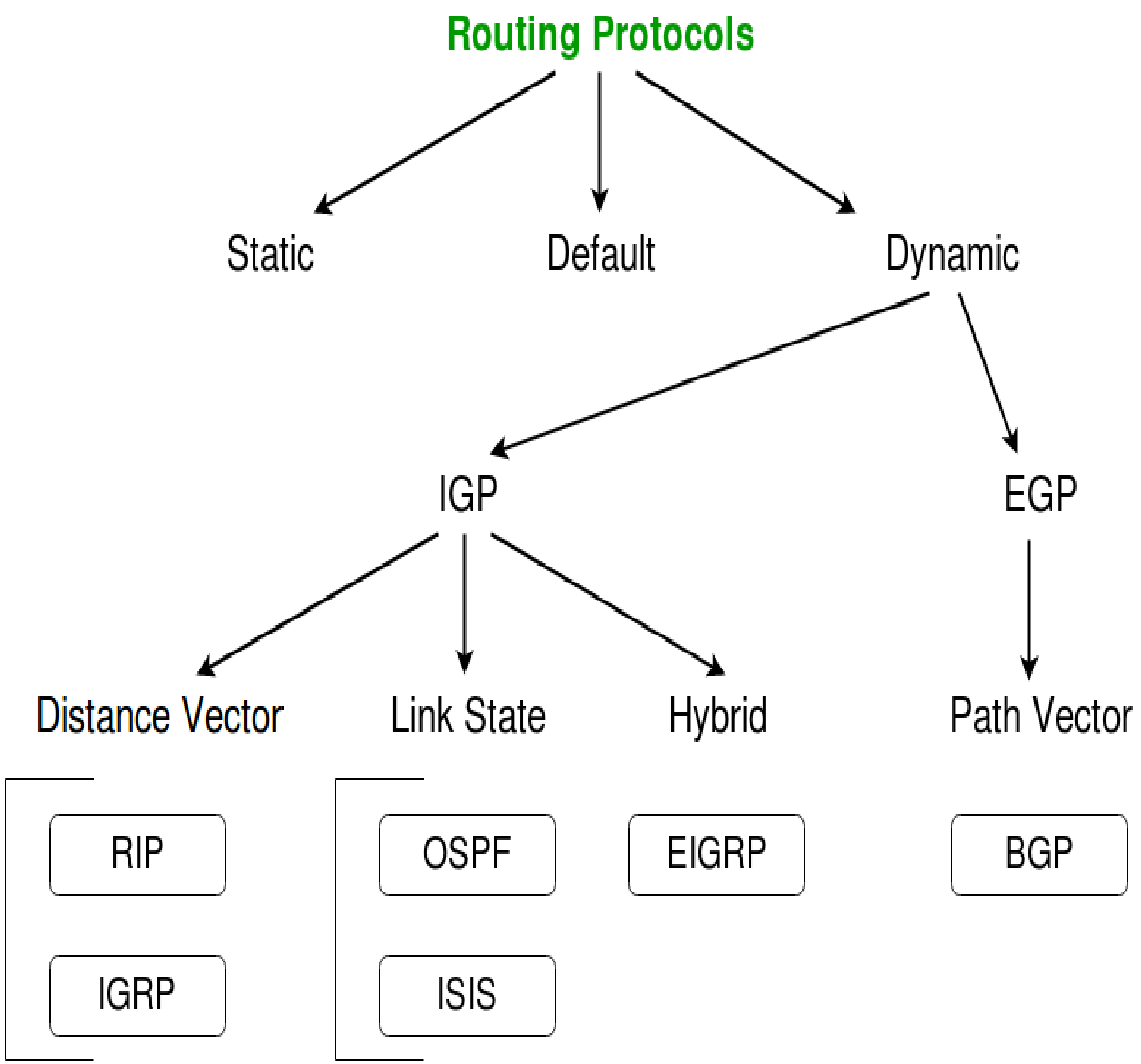
- A **Dynamic Router** (or more precisely, a router that uses *dynamic routing protocols*) is a networking device that automatically **learns**, **updates**, and **shares** information about network routes with other routers in real-time.
- Unlike **Static Routing**, where an administrator manually enters routes, **Dynamic Routing** allows routers to **discover** and **adapt** to network changes automatically (like new routers being added or links going down).
- **Dynamic routing** is known as a technique of finding the best path for the data to travel over a network in this process a router can transmit data through various different routes and reach its destination on the basis of conditions.


R4
R7
R2
R9
R1
R5
R10
R3
R8
R6


R4
R7
R2
R9
R1
R5
R10
R3
R8
R6

How Dynamic Routing Works

- Router Discovery:Each router identifies its neighbors using a routing protocol (like RIP, OSPF, or EIGRP).
- Exchange of Routing Information:Routers share information about the networks they know.Example: Router A tells Router B, “I can reach Network X.”
- Building the Routing Table:Each router uses the information from neighbors to build a routing table, which lists all possible paths and the best path to reach each network.
- Automatic Updates:If a route fails (say, a cable breaks), routers detect the change and automatically recalculate routes.


Routing Protocols
Static
Default
Dynamic
IGP
EGP
Distance Vector
Link State
Hybrid
Path Vector
RIP
IGRP
OSPF
ISIS
EIGRP
BGP

Abbreviations -

- **IGP** – Interior Gateway Protocol
  EGP – Exterior Gateway Protocol
  RIP – Routing Information Protocol
  IGRP – Interior Gateway Routing Protocol
  OSPF – Open Shortest Path First
  ISIS – Intermediate System to Intermediate System
  EIGRP – Enhanced Interior Gateway Routing Protocol
  BGP – Border Gateway Protocol

IGP – Interior Gateway Protocol

- **IGP (Interior Gateway Protocol)** is a **type of routing protocol** used **within a single autonomous system (AS)** — that is, **within one organization’s internal network.**
- IGP is used to help routers inside one company or network domain find the best paths to send data packets to each other.

Distance vector routing

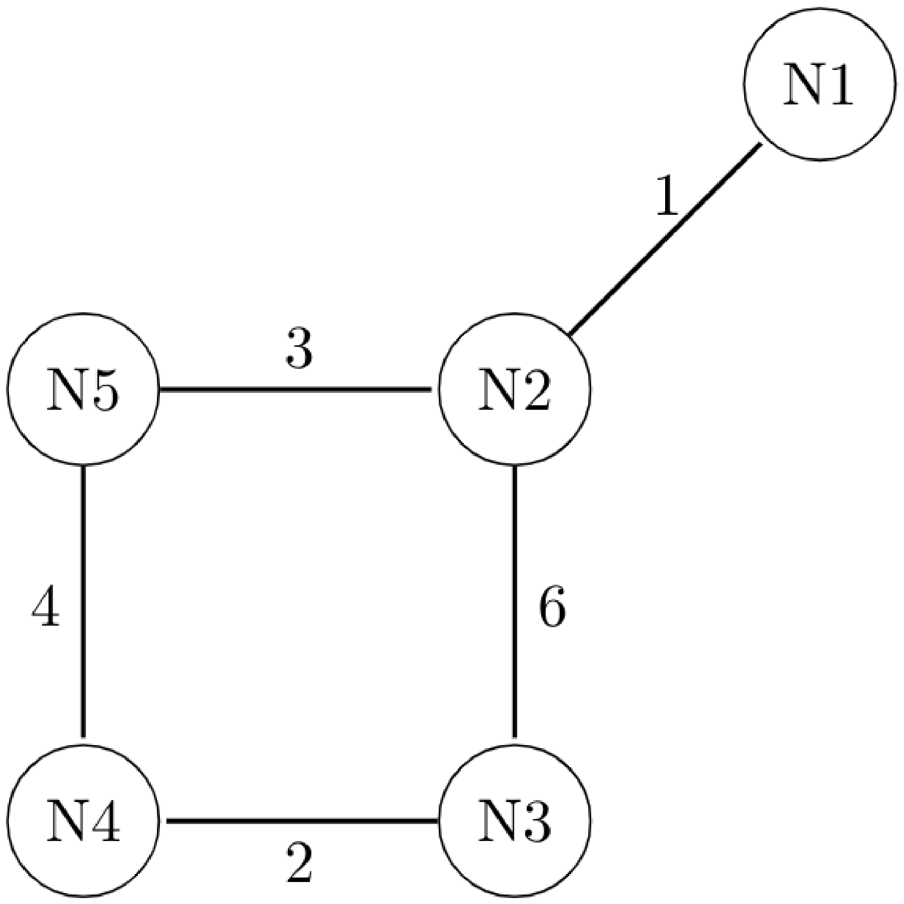
- **RIP** – Routing Information Protocol
  IGRP – Interior Gateway Routing Protocol
- **Distance Vector Routing** is a **dynamic routing method** where routers determine the best path to reach a destination **based on distance (metric)** and **direction (vector)**.

Distance = How far the destination is (often measured in *hops*).

Vector = The direction or the next router (next hop) to send the packet to.

Problem

Consider a network with five nodes,N1 to N5 , as shown as below. The network uses a Distance Vector Routing protocol. Once the routes have been stabilized, the distance vectors at different nodes are as follows.



The network uses a Distance Vector Routing protocol. Once the routes have been stabilized, the distance vectors at different nodes are as follows.

N1: $(0, 1, 7, 8, 4)$

N2: $(1, 0, 6, 7, 3)$

N3: $(7, 6, 0, 2, 6)$

N4: $(8, 7, 2, 0, 4)$

N5: $(4, 3, 6, 4, 0)$

Each distance vector is the distance of the best known path at that instance to nodes, $N1$ to $N5$, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

The cost of link $N2 - N3$ reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, $N3$?

A. $(3, 2, 0, 2, 5)$
B. $(3, 2, 0, 2, 6)$
C. $(7, 2, 0, 2, 5)$
D. $(7, 2, 0, 2, 6)$

Count to infinity problem

- The **Count to Infinity Problem** is a classic issue in **Distance Vector Routing (DVR)** protocols like **RIP (Routing Information Protocol)**. It occurs when there’s a change in network topology—especially a **link or node failure**—and routers continuously increase their distance metric to a destination, never quickly realizing that the destination has become unreachable.

Link state

Internet protocol

- The Internet Protocol (IP) is a set of rules that allows computers and other devices to communicate over the Internet. It ensures that information sent from one device reaches the correct destination by using a unique set of numbers known as IP addresses.
- The Internet Protocol is a fundamental component of the Internet and computer networks, responsible for delivering packets of data from the source host to the destination host based on their IP addresses. It ensures that packets of data get to the right destination from the source device.

Purpose of IP

- The main job of the Internet Protocol is to **identify devices** and **route data** between them. Every device connected to the internet has an **IP address**, which acts like a digital “home address.”

IP Address

- An **IP address** is a unique numerical label assigned to each device connected to a network.
- There are two main versions: IPv4 and IPv6

IPv4

- Next ppt